

Math 118 Lec 8

plan: limits, continuity
convergence of a sequence in a normed space
modes of convergence for sequences of functions $\left\{ \begin{array}{l} f_n \rightarrow f \\ \text{pointwise} \\ \text{uniformly} \\ \text{in } L^p \end{array} \right.$

Recall that a norm measures distance.

def: A function $f: X \rightarrow Y$ is continuous at x_0 if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } \|x - x_0\|_X < \delta \Rightarrow \|f(x) - f(x_0)\|_Y < \varepsilon$$

In words: You can make $f(x)$ as close to $f(x_0)$ as you want (specified by $\varepsilon > 0$) by choosing x close enough to x_0 (specified by $\delta > 0$).

def: $f: X \rightarrow Y$ is continuous if it is continuous at each $x_0 \in X$.

Theorem: A linear operator $T: X \rightarrow Y$ is continuous iff it is bounded.

proof: $\Leftarrow$ Given $\|Tx\| \leq M\|x\| \quad \forall x \in X$.

Must show that T is continuous.

Let $x_0 \in X$, $\varepsilon > 0$. Let $\delta = \varepsilon/M$.

Then if $\|x - x_0\| < \delta$ we have

$$\|Tx - Tx_0\| = \|T(x - x_0)\| \leq M\|x - x_0\| < M\delta = \varepsilon$$

$\Rightarrow$ Suppose T is continuous.

Must show $\exists M$ s.t. $\|Tx\| \leq M\|x\| \quad \forall x \in X$.

Let $\varepsilon = 1$. Since T is continuous at $x_0 = 0$, $\exists \delta > 0$ s.t.

$$\|x - x_0\| < \delta \Rightarrow \|Tx - Tx_0\| < \varepsilon = 1$$

$$\|x\| < \delta \Rightarrow \|Tx\| < 1 \quad (x_0 = 0, Tx_0 = 0)$$

and for any $\tilde{\delta} \in (0, \delta)$, $\|u\| \leq \tilde{\delta} \Rightarrow \|Tu\| < 1 \Rightarrow \|Tu\| \leq 1$.

Suppose $x \neq 0$. Then

$$\|Tx\| = \left\| \frac{\|x\|}{\tilde{\delta}} T\left(\underbrace{\frac{x}{\|x\|}}_u \tilde{\delta}\right) \right\| = \frac{\|x\|}{\tilde{\delta}} \|Tu\| \leq \frac{\|x\|}{\tilde{\delta}}$$

where we used $\|u\| = \tilde{\delta}$ here 

T is bounded and $\|T\| \leq \frac{1}{\tilde{\delta}}$ for all $\tilde{\delta} \in (0, \delta)$.

$$\therefore \|T\| \leq \frac{1}{\delta}.$$

def: We say a sequence $\{u_n\}_{n=1}^{\infty}$ converges to u
if $\lim_{n \rightarrow \infty} \|u_n - u\| = 0$

ε - δ def: $\forall \varepsilon > 0 \quad \exists N$ s.t. $n \geq N \Rightarrow \|u_n - u\| < \varepsilon$

Synonyms: $u_n \rightarrow u$, $\lim_{n \rightarrow \infty} u_n = u$, $\|u_n - u\| \rightarrow 0$

Theorem: A convergent sequence is bounded.

proof: Suppose $u_n \rightarrow u$. Let $\varepsilon = 1$. Then $\exists N$ s.t.

$$\|u_n - u\| < 1 \text{ for } n \geq N.$$

$$\|u_n\| = \|u_n - u + u\| \leq 1 + \|u\| \text{ for } n \geq N.$$

$$\therefore \text{For all } n, \|u_n\| \leq M = \max \left(\max_{1 \leq n < N} \|u_n\|, 1 + \|u\| \right)$$

Theorem: $F: U \rightarrow V$ is continuous at $\tilde{u}$ iff

(from Math 104) $F(u_n) \rightarrow F(\tilde{u})$ for every sequence $u_n \rightarrow \tilde{u}$

$$\left(\text{continuity: } \lim_{u \rightarrow \tilde{u}} F(u) = F(\tilde{u}) \right) \text{ vs } \left(\lim_{n \rightarrow \infty} F(u_n) = F(\tilde{u}) \text{ whenever } u_n \rightarrow \tilde{u} \text{ as } n \rightarrow \infty \right)$$

This means you can take limits through a continuous function.

$$\lim_{n \rightarrow \infty} F(u_n) = F\left(\lim_{n \rightarrow \infty} u_n\right)$$

Example: Show that the inner product is continuous from $V \times V \rightarrow \mathbb{C}$

solution: Suppose $(x_n, y_n) \rightarrow (x, y)$ in $V \times V$. Then

$$\begin{aligned} |\langle x_n, y_n \rangle - \langle x, y \rangle| &= |\langle x_n, y_n \rangle - \langle x, y_n \rangle + \langle x, y_n \rangle - \langle x, y \rangle| \\ &\leq |\langle x_n - x, y_n \rangle| + |\langle x, y_n - y \rangle| \\ &\leq \|x_n - x\| \cdot \|y_n\| + \|x\| \|y_n - y\| \\ &\leq \|x_n - x\| M + \|x\| \|y_n - y\| \rightarrow 0 \text{ as } n \rightarrow \infty \end{aligned}$$

Modes of convergence for sequences of functions

current setting: $f: I \rightarrow \mathbb{C}$, $I = \text{one of}$ $\begin{cases} [a, b] \\ [a, \infty) \\ (-\infty, b] \\ (-\infty, \infty) \end{cases}$

$f_n \rightarrow f$ can mean several things:

1. pointwise convergence ($\forall x \in I, f_n(x) \rightarrow f(x)$)

i.e. $\forall \varepsilon > 0, x \in I \exists N \text{ s.t. } \forall n \geq N, |f_n(x) - f(x)| < \varepsilon$

2. uniform convergence $\uparrow N \text{ depends on } \varepsilon, x$

$\forall \varepsilon > 0 \exists N \text{ s.t. } \forall n \geq N, x \in I: |f_n(x) - f(x)| < \varepsilon$

$\uparrow$ the same N works for all x

3. convergence in $L^p(I)$ ($p=1$ or 2)

$$\lim_{n \rightarrow \infty} \|f_n - f\|_p = 0$$

$p=1$: $\lim_{n \rightarrow \infty} \int_a^b |f_n(x) - f(x)| dx = 0$

$p=2$: $\lim_{n \rightarrow \infty} \sqrt{\int_a^b |f_n(x) - f(x)|^2 dx} = 0$

equivalently, $\lim_{n \rightarrow \infty} \int_a^b |f_n(x) - f(x)|^2 dx = 0$

$$\lim_{n \rightarrow \infty} (c_n)^{\frac{1}{k}} = \left(\lim_{n \rightarrow \infty} c_n \right)^{\frac{1}{k}} = (0)^{\frac{1}{k}} = 0$$

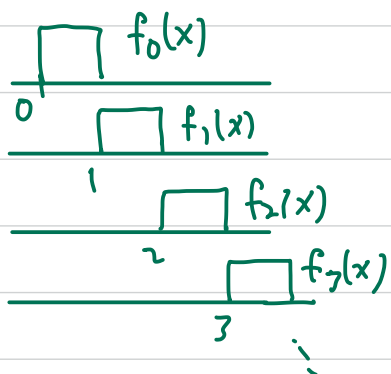
3 is just convergence in the L^p norm.

2 is convergence in the uniform norm (∞ -norm):

$$\|f\|_{\infty} = \sup_{x \in I} |f(x)|$$

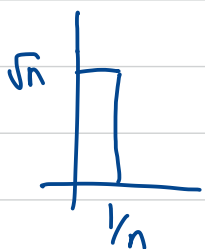
There is no norm associated with pointwise convergence

Examples: ① $I = \mathbb{R}$, $f_n(x) = \begin{cases} 1 & n \leq x < n+1 \\ 0 & \text{o.w.} \end{cases}$



$f_n \rightarrow 0$ (the zero function)
pointwise, but not uniformly
or in L^1 or L^2

② $I = (0, 1)$, $f_n(x) = \begin{cases} n^{1/2} & 0 < x < \frac{1}{n} \\ 0 & \frac{1}{n} < x < 1 \end{cases}$



$f_n \rightarrow 0$ pointwise

$f_n \not\rightarrow 0$ uniformly

$f_n \rightarrow 0$ in L^1

$f_n \not\rightarrow 0$ in L^2

$$\int_0^{1/n} \sqrt{n} dx = \frac{1}{\sqrt{n}} \rightarrow 0$$

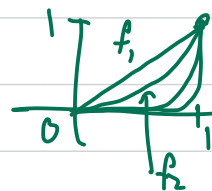
$$\int_0^{1/n} |\sqrt{n} - 0|^2 dx = n \cdot \frac{1}{n} = 1 \neq 0$$

③ $I = (0, 1)$, $f_n(x) = x^n$

$f_n \rightarrow 0$ pointwise (since $1 \notin I$)

$f_n \not\rightarrow 0$ uniformly

$f_n \rightarrow 0$ in L^1 and L^2



$$0.99^n = e^{n \underbrace{\ln(0.99)}_{\text{negative}}} \xrightarrow{\text{as } n \rightarrow \infty} e^{-\infty} = 0$$

$$\textcircled{4} \quad I = \mathbb{R}, \quad f_n(x) = \begin{cases} 1/n & 0 \leq x < n \\ 0 & \text{o.w.} \end{cases}$$



$f_n \rightarrow 0$ pointwise, uniformly, in L^2 , but not in L^1

Theorem: If $f_n \rightarrow f$ uniformly and $I = [a, b]$ is a bounded interval, then $f_n \rightarrow f$ in L^1 and L^2 .

proof

L^1 : Let $\varepsilon > 0$. Pick N so $n \geq N \Rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{b-a}$.

$$\begin{aligned} \text{Then for } n \geq N, \quad \|f_n - f\|_{L^1} &= \int_a^b |f_n(x) - f(x)| dx \\ &\leq \int_a^b \frac{\varepsilon}{b-a} dx = \varepsilon \end{aligned}$$

L^2 : Let $\varepsilon > 0$. Pick N so $n \geq N \Rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{\sqrt{b-a}}$

$$\begin{aligned} \text{Then for } n \geq N, \quad \|f_n - f\|_{L^2} &= \sqrt{\int_a^b |f_n(x) - f(x)|^2 dx} \\ &\leq \sqrt{\int_a^b \frac{\varepsilon^2}{b-a} dx} = \sqrt{\varepsilon^2} = \varepsilon \end{aligned}$$